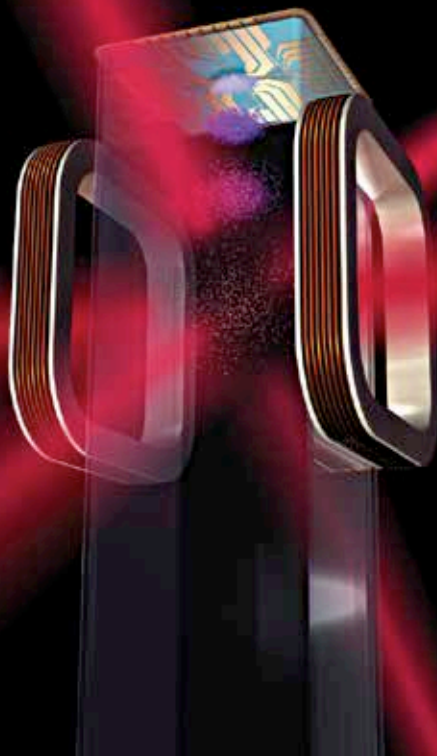


## Space age

Join humanity's age-long quest to figure it all out – one particle of exotic matter at a time.

## Speaker for the dead

Facebook has given a father full access to his dead son's social media page and its contents. Does this raise privacy concerns?



NASA's Cold Atomic Lab Logo

# On the edge

Exploring our fascination with the extreme

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**T**ime and time again we are reminded of the limits of our universe and turn our backs to it, only to have one enterprising research institute pat our shoulder and ask us to see what lies beyond. Every new limit keeps getting overwritten by a new one, and impossible seems to be a statement you would not want to make in public lest you be made a fool of in a short

period of time. This is good. This is exactly how it should be.

## The Coldest

The coldest spot on the earth right now is a ridge in the Antarctic where the temperature can dip to 92 degrees Celsius. The coldest spot in the universe is the Boomerang Nebula at -272 degrees Celsius or 1K, a cloud of gas released out by a dying star around 5,000 light-years away. NASA researchers spit on such ridges and Nebula's and intend to create the coldest

spot in the Universe at 100 picoKelvin, or one ten billionth of a Kelvin. And that too inside the International Space Station.

In what has been imaginatively called (cold?) the Cold Atom Lab, was put together to provide scientists with a doorway into the world of quantum mechanics. As temperatures start plummeting to near zero Kelvin, or Absolute Zero, strange things start happening. Molecules don't react the way they normally do, two atoms may combine to result in no atom at all – a world of probability and far-fetched science-fiction films is on the horizon and scientists are all too keen on finding out what happens there. And if what they believe is correct, we may be able to see quantum wave packages as wide as a human hair.

But why on earth will this be on a space-ship? Well consider a spray can. It cools when used because the gas inside the can leaves, the remaining gas expands to fill in that space and as a rule of thermodynamics, gases cool when they expand. To create the coldest spot, researchers have to use a more sophisticated instrument mirroring our humble spray can called 'magnetic traps'. In the